Barcode & Digital Document Management System

1. Meeting Overview

Project: Barcode/QR Code-Based Document Tracking & Digital Document Management System
Organizations: GCTL Infosys & Formosa Logistics Limited
Key Participants: Muhammad Abdul Mannan, Humayun, Mohammad Kayum Ali Shikder, Sifat Ahmed, Farhad Hossain, Md. Samaul Islam, Nazmul Hoque, Ray

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2. Key Decisions & Requirements

Document Management

- Multiple documents can be uploaded against a **Job Number** and **Bill of Entry (B/E) Number**.
- Supported file types: **PDF, JPEG, PNG**.
- Maximum file size: **15 MB**.
- Upload fields:
    - Document Name
    - Document Type
    - Description
    - Others / Custom Notes
- System-generated information:
    - File Size
    - File Type
    - Document Code
    - Upload Date & Time
    - User ID

Barcode & QR Code

- Linear Barcode: **Code 128**
- 2D Code: **QR Code**
- Barcode format: `DOC + YYYY + MM + 8-digit Serial`
- Serial number will reset annually.
- Barcode label will contain:
    - Formosa Logistics Ltd.
    - Barcode/QR Code
    - Job Number
    - Document Reference

    - Timestamp
    - User ID

Master Summary Cover

- A **single-page Master Summary Cover** will be generated for each physical file.
- The cover will include:
    - Job Number
    - B/E Number
    - Document Checklist
    - Master Barcode/QR Code
- Scanning the Master Barcode/QR Code will open the complete digital document dossier.
- Individual document barcode printing will remain available through an optional checkbox.

Search & Reference

- **B/E Number** will be the primary external reference for Customs, VAT, NBR and Bond-related queries.
- Documents must be retrievable using either **Job Number or B/E Number**.

Physical Document Retention

- Physical documents containing official customs clearance seals must be retained for **5 years** for audit purposes.
- Final digital document upload will be completed after physical customs clearance and delivery verification.

External & Client Access

- Clients and branches will access documents through:
    - QR Code scanning
    - Secure portal/application login
    - Role-based access permissions
- No complex third-party API middleware is planned.

Mobile Application

- **Android:** Production application is published and active on Google Play Store.
- **iOS:** Application has been submitted to the Apple App Store and is awaiting approval.

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3. Implementation Phases

Phase 1 — Physical File & Master Barcode

- Standardize Job/B/E numbering.
- Generate and print Master Barcode/QR Code cover sheets.
- Implement physical file transit tracking.

Phase 2 — Digital DMS & Checklist

- Implement shipment-specific checklists for:
  - Air Cargo
  - Sea Freight
  - Export
- Enable digital document upload.
- Track missing and pending documents.

Phase 3 — Management Dashboard & Control

- Real-time operational dashboard.
- Aging analysis for unbilled and pending jobs.
- Employee-wise pending reports.
- Automated alerts and notifications.

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4. Action Items

GCTL Infosys

1. Prepare the complete metadata structure and document checklist in a master **Excel template**.
2. Define the Summary Cover Sheet layout and required fields.
3. Add **Others / Custom Notes** functionality for non-standard NOCs and clearances.
4. Share the Excel template with Formosa Logistics for review.
5. Monitor and update the iOS App Store release status.

Formosa Logistics

1. Review the Excel template with the **Dhaka and Chittagong operational teams**.
2. Finalize mandatory document checklists for:
   - Air Cargo
   - Sea Freight
   - Export
3. Provide feedback and confirm the finalized requirements to the GCTL Infosys team.